# What changed after the submission

*A running record of every change to NeuroMPS after the round-1 deadline (2026-10-10 15:59:59 UTC; the submitted state is commit `464eb76`). Each entry opens with the one sentence a slide would carry, then says what changed, why, what it did to the results, and where the evidence is. Numbers cite `outputs/metrics.json` as the report does, so the citation check holds this document to the same standard. Newest entries are at the end; planned work is in the queue at the bottom.*

| # | Date | Change | Effect on results | Commits |
|---|---|---|---|---|
| **1** | 2026-10-10 | Unused Kaggle notebook removed; seven figures redrawn at print size | none (no number or claim changed) | `e5ed404`, `bbd75f9` |
| **2** | 2026-10-11 | Bicuculline doses put on one unit scale | the dose conclusion reverses | `886d8d3` |
| **3** | 2026-10-22 | Demo app runs as a hosted page from the repository alone | none; access only | `c4a9004` |
| **4** | 2026-10-22/23 | From chip to animal: the drug knobs on a whole fly brain | new results, pre-registered | `c83eb14` … `aba184c` |
| **5** | 2026-10-24 | Report page limit lifted; the fly study joins the report | none; structure only | `1c7d94f` |
| **6** | 2026-10-24 | EPA screen cutoffs set beside the fly's precision needs | new comparison (descriptive) | `cd074b3` |
| **7** | 2026-10-24 | Vesicle depletion tested as the cause of the GABA-A reversal | registered prediction failed; new negative result | `3071ab4`, `8fb5065` |
| **8** | 2026-10-24 | The twin as a source of labelled data, and as a prior for the classifier | 1 of 3 registered predictions pass | `761068e`, `93bb4f6` |
| **9** | 2026-10-24 | Electrodes on the fly brain's output tract against a planar array | 4 of 4 registered predictions pass | `cd074b3`, `2c66067` |
| **10** | 2026-10-24 | Behaviour read at the motor neurons of the male CNS | 6 of 7 registered predictions pass | `a1c5f27`, `a2f4ee2` |
| **11** | 2026-10-24 | The motor-level ties resolved on a finer dose grid | 1 of 2 registered predictions pass; one tie resolves | `69db4e6`, `796a33b` |
| **12** | 2026-10-24 | Roadmap to an embodied, closed-loop fly | plan, no new results | `42ee520` |
| **13** | 2026-10-24 | Embodied stage 1: the male CNS drives a muscle-driven leg | both registered tests fail; finding the cause changes the plan | `cbdd6d6` … this branch |
| **14** | 2026-10-24 | Embodied stage 1c: the male nerve cord from descending neurons, brain removed | 3 of 5 registered predictions pass | `739f6f0` … `e576cd9` |
| **15** | 2026-10-24 | Embodied stage 1d: a held reach against a rhythmic swing, fresh seeds | 4 of 4 registered predictions pass | `66502eb` … `19ab0ae` |
| **16** | 2026-10-24 | Embodied stages 1e and 1f: the FlyWire brain drives the male nerve cord and the leg | 1e uninterpretable as registered; 1f 5 of 5 pass on fresh seeds | `f2f5ad3` … this branch |

### 1. Housekeeping pushed just after the deadline

**Slide message:** none; listed for completeness.

Two commits reached GitHub after the deadline: one removed a Kaggle notebook nothing used (`e5ed404`), the other redrew seven figures at the size they print at (`bbd75f9`). They were reverted during round-1 judging so that reviewers saw the submitted state (`2ecca42`), and restored after it (`a05b013`).

### 2. A unit error hid a dose trend, and another team's result exposed it

**Slide message:** *Burst rate falls with bicuculline dose after all; we had ranked nanomolar doses as if they were micromolar, and found the bug by taking a competitor's disagreement seriously.*

- **What changed.** ScreenNeuroPharm records its two lowest bicuculline doses in nM and the rest in µM. Three analyses ranked the dose value without its unit, so 0.1 µM sat among the highest doses. One function, `models/dataset.doses_in_um`, now converts in all of them, and a test (`tests/test_dose_units.py`) checks each.
- **Why.** NeuroChip Twin, another entry, reported a bicuculline dose trend that our report said did not exist. Tracing the disagreement found the bug.
- **Effect.** The feature dose test moved from no feature tracking dose to 2 of 7: burst rate (Spearman −0.5813) and network-burst rate (−0.604) fall with dose across 113 recordings. The twin's inferred inhibition still does not follow dose, so the conclusion became: the measurement carries the trend, and the inverse model is what fails.
- **Evidence.** Report §18 (negative results) and Appendix M; `src/neuromps/models/dataset.py`.

### 3. The demo runs as a hosted web page

**Slide message:** *Judges can open the demo in a browser; nothing to install.*

- **What changed.** Streamlit Community Cloud installs a requirements file and runs `app/app.py` without installing the package. The app now finds `neuromps` in `src/`, and `app/requirements.txt` pins only what the page needs, at the project's versions (a test keeps them equal). The one live-simulation button says it needs the full local install instead of failing.
- **Effect.** None on results. Checked in a clean environment with only the hosted requirements: every tab renders.
- **Evidence.** `tests/test_app_hosted.py`.

### 4. From chip to animal: the chip's drug knobs on a whole fly brain

**Slide message:** *On a simulated whole fly brain, the chip's firing rate ranks drugs correctly but raises the alarm late: in every drug family, behaviour failed first.*

- **What changed.** A new package, `src/neuromps/fly/`, runs the connectome-constrained model of Shiu et al. (2024) on FlyWire with 138,639 neurons, applies the twin's drug knobs by transmitter class, and reads the brain both as a chip (a virtual electrode array scored by `features/mea.py`) and as an animal (the neurons the published model validated against feeding, aversive gating and grooming). It is replicated on the male CNS connectome (165,122 neurons). New data sources, all CC-BY or MIT, enter the manifest as optional; `run_all.py --stage fly` runs it, and the app has a fourth tab.
- **Why.** Round 1 scored us lowest on practical value. A chip assay is used to predict what a drug does to a nervous system, which no culture can check; a mapped, simulated brain can.

- **How it was kept honest.** Three pre-registrations, each committed before the simulations it scores; the release of every table checked against the model's (the published neuron lists come from an older release); every failed prediction reported.
- **Effect.**
- The model is the published one: sugar drives the feeding motor neuron at 76.75 Hz against 77.375 Hz on the release the paper used.
- The chip's firing-rate change tracks behavioural disruption (Spearman 0.863; 0.878 on the male CNS).
- Registered prediction failed: the chip, at the registered threshold, flags the drug after behaviour has failed in all five drug families.
- At behavioural onset, GABA block has moved firing rate by 0.149 log2, about the untreated variation of the best real chips in the asset (0.156).
- A candidate early-warning feature (network-burst rate) proved to be noise: untreated recordings already cross its threshold in a fraction 0.382 of splits.
- Spontaneous activity recovers the direction of wiring between brain regions at chance (0.4946).
- **Evidence.** Report Act VII (§23–§26); `preregistration/`; figures `outputs/figures/fly_*_SIMULATED.png`; app tab 4.

### 5. The report has no page limit; the fly study joins it

**Slide message:** none; structure only.

- **What changed.** The 15-to-20-page gate on the report's main matter is removed from the build and from CI, now that the round-1 page limit no longer applies; a lower bound remains so a broken build that drops sections is still caught. The fly study moves from a companion note into the report as Act VII (§23–§26), after §22 so that no earlier section number changes; the short Appendix P that pointed to the note is gone.

### 6. A regulatory screen's hit cutoff, beside what the fly says a chip must see

**Slide message:** *EPA's MEA screen calls a firing-rate hit only beyond a change about six times larger than the one at which the fly's behaviour had already failed.*

- **What changed.** `src/neuromps/fly/screen_cutoffs.py` reads the hit cutoff of the ToxCast MEA firing-rate endpoints from the asset's 20,375 ToxCast records and converts it to log2. The acute screen normalises activity to a positive control at −100 (activity abolished); its cutoff of 70.97 means suppression to about 29% of control, a change of −1.784 log2, over 515 screened chemicals.
- **Why.** The review pointed out that most of the asset's records sat outside the modelling. This puts them to a use the fly study makes possible: does the screen's own threshold leave room to catch the mechanisms in time?
- **Effect.** The largest firing-rate change at which the fly's behaviour had already failed is 0.304 log2 (acetylcholine block), so the screen's cutoff is 5.87 times larger. Caveat: the cutoff applies to a fitted curve's maximum response across the tested range, not to one dose, so this compares magnitudes, not decisions. The fly side is simulated; the cutoffs are real.
- **Considered and dropped.** A compound-by-compound potency bridge between our spike-level sources and ToxCast: only four compounds appear in both (bicuculline, deltamethrin, acetaminophen, glyphosate), too few to rank.

### 7. Vesicle depletion does not explain why real chips slow down under GABA-A block

**Slide message:** *We tested the reviewer's best explanation for our twin's one wrong direction, registered in advance, and it does not hold: depletion shrinks the firing-rate rise but never turns it into the fall real chips show.*

- **What changed.** `src/neuromps/twin/depression_drugs.py` runs the four drug knobs and a five-step GABA-A dose series under each depression setting (Tsodyks–Markram depletion at the recurrent excitatory synapses), and scores a rule registered before the runs (`preregistration/twin_depression_under_block.json`).
- **Why.** The post-round-1 review proposed depletion as the missing mechanism behind three weak points at once: the twin's wrong firing-rate direction under GABA-A block (§14), its missed bicuculline dose trend (§18) and the cross-platform reversal (§11).
- **Effect.** Under strong GABA-A block the twin's firing rate rises by 2.389 log2 without depletion and by at least 0.8687 with it; real GABA-A antagonist pairs change by −0.335. None of the 7 settings qualifies, and burst rates do not fall with dose under any. Registered outcome: V1 fail, V2 pass, V3 not applicable. The weights were not recalibrated with depletion in place, so the mechanism is not ruled out, only depletion added to the present twin.
- **Evidence.** Report §18, row 9 and figure; `outputs/twin/depression_under_block_simulated.png`.

### 8. Trained on the twin alone, the classifier recognises real mechanisms

**Slide message:** *A classifier that never saw a real label recognises real drug mechanisms at two and a half times chance; the same simulations do not improve a classifier trained on real labels.*

- **What changed.** `src/neuromps/models/twin_prior.py` simulates 30 labelled pairs for each of the five mechanism classes the twin has a knob for, and tests two uses of them, registered before simulating (`preregistration/twin_prior_classifier.json`): as the centre of the classifier's penalty (the review's Fix 2), and as the only training data.
- **Effect.**
- Trained on simulated pairs alone and applied to the 73 real pairs: balanced accuracy 0.41, interval 0.2808 to 0.5072, against one in six by chance (pass).
- As a prior: balanced accuracy 0.53 against the baseline's 0.537, and conformal sets unchanged (two fails).
- A finding for free: real sodium-channel blockers are never recognised, because the twin's threshold knob moves firing by −3.273 log2 where the real blockers move it by −0.795: right direction, wrong size.
- **Evidence.** Report §14, last paragraph.

### 9. Record the tract, not the tissue around it

**Slide message:** *On the simulated fly brain, electrodes on the output tract (the brain's own microchannels) catch GABA block before behaviour fails; a planar array of the same size does not.*

- **What changed.** `src/neuromps/fly/tract.py` reads the parent sweep's spontaneous recordings through three arrays: the full planar array, 64 of its electrodes, and 64 single-unit electrodes on descending neurons. The test, at one threshold set by real-chip noise, was registered before recording (`preregistration/fly_output_tract_electrode.json`).
- **Why.** The review's Investigation B: does instrumenting the connection between compartments reveal failure a flat array misses? Our real devices (§15) had already said that only electrodes inside microchannels see direction.

- **Effect.** All four registered predictions pass. Under GABA block, at behavioural onset, the tract's firing rate has moved by 0.361 log2 against 0.044 for the equal-count planar array; the tract flags 3 families in time against 2, with no false alarms. Checks added after the pass: over ten random draws of each array the tract is ahead in 7 of 10, and the descending-neuron behaviour readout is not what drives it.
- **Evidence.** Report §24 ("Where to put the electrode") and figure; §20, fifth area.

### 10. One step closer to behaviour: the motor neurons

**Slide message:** *Read at the motor neurons that move the fly's mouthparts and legs, the chip is never ahead of behaviour: it is late in three drug families and level in two.*

- **What changed.** `src/neuromps/fly/motor.py` repeats the male-CNS replication's behaviour trials (same seeds) and reads them at the proboscis motor pool (feeding and its bitter gate), the front-leg motor pool (antennal grooming) and all motor neurons (recruitment). Registered before any drug condition was run (`preregistration/fly_motor_readout.json`).
- **Why.** The behaviour readouts were firing rates of interneurons; the male connectome includes the nerve cord, so behaviour can be read where it leaves the nervous system. This is the first step toward a closed-loop, embodied fly, which needs a body and sensory feedback on top.
- **Effect.** Six of seven predictions pass. The interneuron readouts reproduce exactly; sugar drives the proboscis pool at 31.405 Hz and bitter cuts it to 5.625; the chip tracks motor disruption (Spearman 0.8922). The chip-is-late prediction failed narrowly: late in 3, tied in 2, earlier in 0 (four late were needed). Motor output fails no later than the interneurons in every family.
- **Evidence.** Report §25 ("One step closer to behaviour") and figure; `outputs/fly/malecns_motor_SIMULATED.csv`.

### 11. Resolving the two ties, without re-scoring the failed test

**Slide message:** *We did not move the goalposts on the failed test; we ran a new registered one. With the ties resolved, the chip is late in four drug families, level in one, and never ahead.*

- **What changed.** `src/neuromps/fly/tie_break.py` runs four finer dose levels inside each of the two tied steps of entry 10, with the same seeds and untreated references, registered before running (`preregistration/fly_motor_tie_break.json`).
- **Why.** Entry 10's chip-is-late test failed because two families tied, both onsets inside one coarse dose step. Changing the test after the fact would have undone the registration; a new, registered test with a finer grid answers the open question honestly.
- **Effect.** The re-run tied levels reproduce exactly. The raised threshold resolves to chip later (motor at 0.32, chip at 0.4); GABA potentiation stays tied. The follow-up predicted both would resolve, so it failed. Entry 10's test stays failed; combined, the chip is later in 4, level in 1, earlier in 0.
- **Evidence.** Report §25 and figure; `outputs/fly/malecns_motor_tie_break_SIMULATED.csv`.

### 12. A roadmap, not a rushed demo: the embodied, closed-loop fly

**Slide message:** *Next, the simulated fly gets a body and its sensory feedback, so that "does the chip flag a drug before behaviour fails?" can be asked of movement itself; six stages, each with a registered go/no-go test.*

- **What changed.** `report/ROADMAP_EMBODIED_FLY.md` (Word and PDF beside it) sets out the plan: brain and nerve cord, motor neurons to muscles, muscles to joints in a physics body (NeuroMechFly v2 on MuJoCo), and joints back to proprioceptive neurons. A small script,

`src/neuromps/fly/roadmap_inventory.py`, counts what the male connectome already provides: of 381 leg motor neurons, 260 are typed by the muscle they drive, and 1,030 proprioceptive sensory neurons can carry the feedback.

- **Why.** Building it properly takes weeks; showing a half-working body would cost credibility. A staged plan with registered tests shows where the project goes and how it will be checked.
- **Evidence.** `report/ROADMAP_EMBODIED_FLY.md`.

### 13. The leg moves, but not specifically: what stage 1 found

**Slide message:** *Wired to a muscle-driven leg, the male fly brain moves it toward the head under antennal drive, but under sugar drive too; the cause is that the published model's parameters do not carry over to the second connectome, where activity avalanches.*

- **What changed.** MuJoCo (an optional dependency) runs the FlyMimic leg from FlyGym 2; `src/neuromps/fly/embodied_stage1.py` maps male-CNS motor neurons onto its muscles by name and drives it open loop. Two registered tests: stage 1 and stage 1b (weight rescaled by the connectomes' input-synapse ratio).
- **Effect.** Stage 1 fails on specificity (sugar brings the claw closer than antennal drive); stage 1b's feeding check fails (sugar drives the feeding motor neuron at 0.0 Hz), so its leg results are uninterpretable. The cause: a sugar drive activates 251 neurons in the published FlyWire model and 14,263 in the male CNS.
- **What it changes.** The next step drives the male nerve cord from its descending neurons alone, then joins the validated FlyWire brain to it (roadmap §7).
- **Evidence.** `report/ROADMAP_EMBODIED_FLY.md` §7; `outputs/fly/embodied_stage1*_SIMULATED..`

### 14. The grooming neurons move the right leg

**Slide message:** *Driven from the descending neurons that start antennal grooming, the male nerve cord moves the front leg, and only the front leg, to the head and holds it there; walking neurons swing it instead.*

- **What changed.** `src/neuromps/fly/embodied_stage1c.py` cuts the male CNS to its nerve cord (brain removed), drives named descending neurons at 150 Hz, and moves the FlyMimic leg as in stage 1. Registered before running.
- **Effect.** Three of five predictions pass: silence without drive; the left front-leg motor pool at 15.846 Hz under grooming drive against 2.72 and 1.697 for the other legs; no other descending neuron recruits the front leg as strongly. The kinematic prediction failed: forward-walking drive dips nearly as close to the head (1.154 mm against 1.1345) at the bottom of a rhythmic swing. A sustained reach and a swing need a different readout, to be registered as stage 1d.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage1c_SIMULATED..`

### 15. The fly's grooming neurons hold its leg at its head

**Slide message:** *On fresh seeds and with readouts registered in advance: driving the fly's antennal-grooming neurons brings the simulated front leg to the head and holds it there for nearly the whole second; walking neurons swing it, escape neurons extend it.*

- **What changed.** `src/neuromps/fly/kinematics.py` adds two readouts (time near the head; rhythmicity), `src/neuromps/fly/embodied_stage1d.py` reruns stage 1c's design on fresh seeds. The readouts were suggested by stage 1c's traces, so they were registered before this run (`preregistration/embodied_stage1d_reach_versus_swing.json`) and tested out of sample.

- **Effect.** All four predictions pass: time near the head 0.9665 under grooming drive against at most 0.368 under any other; forward walking more rhythmic than grooming; the same at every activation scale; the front-leg specificity of stage 1c replicates. Stage 1c's failed predictions stay failed.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage1d_SIMULATED..`

### 16. From an antennal touch in the brain to a leg at the head

**Slide message:** *Antennal input to the validated FlyWire brain, relayed through matched descending neurons into the male nerve cord, moves a muscle-driven leg to the head; sugar input does not, and removing four grooming neurons abolishes the movement.*

- **What changed.** `src/neuromps/fly/model.py` can replay recorded spike trains into chosen neurons; `src/neuromps/fly/relay.py` pairs descending neurons across the two connectomes by cell type and side and joins the FlyWire brain to the male nerve cord; stage 1e and stage 1f were each registered before running.
- **Effect.** Stage 1e's absolute precondition failed (grooming descending neurons at 8.531 Hz against a 10 Hz line), so its leg results are uninterpretable. Stage 1f, on fresh seeds, passes all five predictions: claw to 0.5369 mm under antennal input against 0.9377 under sugar; back to 1.4793 mm when aDN1 and aDN2 are removed.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage1f_SIMULATED..`

### 17. The nerve cord alone does not walk

**Slide message:** *Driven by its walking command, the connectome-only nerve cord fires the leg extensors tonically and leaves the flexors silent: no rhythm. Rhythm needs intrinsic currents the connectome does not record; adding them is the next registered step.*

- **What changed.** `src/neuromps/fly/fictive.py` runs stage 2a (registered before running): the six legs' tibia flexor and extensor pools under walking, grooming and backward-walking commands, phases measured between pools. `src/neuromps/fetch/flybody.py` fetches the flybody whole-body model (Apache-2.0) for the feeding and walking body, each file checked against its git hash.
- **Effect.** Silent without drive (W0 passes); under the walking command the left middle-leg extensors fire at 54.12 Hz and the left front flexors at 0.0 Hz, so there is no rhythm and no phase (W1 to W4 fail). The roadmap listed this risk and the response to it.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2a_SIMULATED..`

### 18. Sugar extends the proboscis, bitter holds it back

**Slide message:** *Sugar input to the validated FlyWire brain drives the motor neurons of each proboscis muscle the literature names, and a simulated proboscis extends fully; bitter input holds it back. Six of six registered predictions pass.*

- **What changed.** `src/neuromps/fly/proboscis.py` drives the flybody proboscis (body fixed, 0.01 ms step, checked against a step half that size); `src/neuromps/fly/embodied_stage2b.py` turns the proboscis motor pools (MN9, MN1/MN2, MN4, MN3, named by FlyWire cell type after McKellar et al. 2020) into rostrum and haustellum commands. Registered before running.
- **Effect.** Full extension under sugar (0.3431 mm of 0.3428); none under bitter, antennal input or none; time-averaged extension from 0.3054 to 0.0488 mm when bitter is added. Sugar recruits only feeding motor neurons (exploratory).
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2b_SIMULATED..`

### Work queue (decided 2026-10-24, from the post-round-1 review)

| Item | Status | Registered before running |
|---|---|---|
| **Synaptic depression under GABA-A block** | done, entry 7 (failed as registered) | yes |
| **ToxCast potency bridge** | replaced by entry 6 (only four shared compounds) | — |
| **Output-tract electrode** | done, entry 9 (4 of 4 pass; 7 of 10 draws) | yes |
| **Brain and nerve cord as a two-chamber chip (male CNS)** | scoped out: one chamber pair is an anecdote, and FlyWire already tested 698 region pairs (§25) | — |
| **Twin posteriors as classifier features** | done as a twin-fitted prior, entry 8 (the delta posterior infers one knob, outside Task 1's domain) | yes |
| **Random-graph-to-connectome blend for organoids** | dropped | — |
| **Embodied, closed-loop fly (body, nerve cord to muscles, sensory feedback)** | roadmap after the final (`report/ROADMAP_EMBODIED_FLY.md`, entry 12); its first step, motor-neuron readouts, is entry 10 | — |

Why the blend was dropped: grafting fly wiring onto a model of a mammalian cortical culture has no biological basis, and organoid–monolayer differences have many confounds, so a match would prove nothing. The embodied fly is not dropped but scheduled: it adds new information only with the loop closed (nerve cord to muscles, body, sensory feedback), which is a research project for after the final; reading the motor neurons (entry 10) is its first step.